

C979

Science Methods—Secondary Physics

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

Students in a class that meets right after lunch test whether ramp height affects how far a toy car rolls. In trial 2 they raise the ramp and also switch to a heavier car. The school just bought new meter sticks. What is the flaw?

- A. Two variables changed, so the effect of ramp height cannot be separated from the effect of mass.
- B. Only one trial per height was run, so no conclusion is ever possible.
- C. The new meter sticks may be inaccurate.
- D. Distance should be the independent variable.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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2. [TRANSFER — reversed stem / different variable]

A student wants to test whether pendulum length affects period. Which variables must be held constant?

- A. Length and period.
- B. Bob mass and release angle.
- C. Release angle only.
- D. Period only.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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3.

A physics teacher who coaches the robotics team shows a puck sliding across a new air-hockey table. A student says, “It keeps moving because the force from the push is still in it.” What is the best teaching response?

- A. Explain how the robotics team’s motors push robots.
- B. Ask students to predict and then compare the puck’s motion on the air table vs. on carpet, and discuss that friction, not a stored force, changes motion.
- C. Agree that the push is stored as force inside the puck.
- D. Give the definition of inertia and move on.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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4. [TRANSFER — reversed stem / different variable]

A cart rolls at a constant 2 m/s on a level track with negligible friction. What is the net force on the cart?

- A.** 2 N forward.
- B.** A small forward force, always needed to keep anything moving.
- C.** A forward force equal to its speed.
- D.** Zero.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

In a lab using carts from a donated equipment kit, cart A (1 kg) moving at 3 m/s collides with cart B (2 kg) at rest on a low-friction track, and they stick together. A student says momentum is not conserved because cart A slowed down. The track is 2 meters long. What is the best response?

- A.** The pair moves at 3 m/s, because momentum is conserved.
- B.** The student is right; cart A lost momentum, so momentum is not conserved.
- C.** The pair moves at 1.5 m/s, spread over the 2-meter track.
- D.** Define the system as both carts; its momentum stays 3 kg·m/s, so the pair moves at 1 m/s.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Using a motion sensor bought with this year's budget, students record a cart's position: 0.5 m at 0 s and 2.5 m at 4 s, in a straight line on a position-time graph. The cart is painted red. What is the cart's velocity?

- A.** 2.0 m/s
- B.** 0.5 m/s
- C.** 2.5 m/s
- D.** 0.625 m/s

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A velocity–time graph for a runner is a straight line from 0 m/s at 0 s to 6 m/s at 3 s. What does the graph give for acceleration and for displacement?

- A.** 9 m/s² and 2 m.
- B.** 6 m/s² and 3 m.
- C.** 2 m/s² and 9 m.
- D.** 2 m/s² and 18 m.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

Teams will design a device to protect an egg dropped from 3 meters. Each team may use \$5 of materials and has two class periods. The winning team last year used bubble wrap. What should students do first?

- A.** Define the criteria for success (egg unbroken) and constraints (\$5, two periods, 3 m drop) before sketching solutions.
- B.** Pick the one correct design from the teacher.
- C.** Start building right away and fix problems after the first drop.
- D.** Buy bubble wrap, since it worked last year.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A team lists “the bridge must hold 10 kg” and “we may use only 50 craft sticks.” Which is the constraint?

- A.** Neither; constraints are only about money.
- B.** Holding 10 kg.
- C.** Using only 50 craft sticks.
- D.** Both are criteria.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A teacher who recently attended a technology conference is choosing a way for students to study a ball's motion in free fall. The fall lasts about half a second. Which choice best serves the learning goal?

- A.** A virtual-reality headset shown at the conference.
- B.** A simulation that shows the answer so students skip the analysis.
- C.** Video analysis or a motion sensor, which capture many position points so students can analyze the graph.
- D.** Hand stopwatches, because technology should be avoided in physics.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Students are building simple series circuits with bulbs in a room that was recently rewired. One group says a wire and battery have become hot after they connected the wire straight across the battery. What should the teacher do?

- A.** Add a second battery so the bulb will light.
- B.** Let them continue, because battery circuits are always safe.
- C.** Tell the group to plug into a wall outlet instead, since the room is newly wired.
- D.** Have the group disconnect the circuit, let it cool, and explain that it was a short circuit.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

Students launch foam balls with a spring launcher to study projectile range. No chemicals are used, and the lab is in the gym for more space. Who needs eye protection?

- A.** Everyone in the launch area, with impact-rated eye protection.
- B.** No one, because no chemicals are used.
- C.** No one, because the gym gives more space.
- D.** Only the student operating the launcher.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

~~Students in a class that meets right after lunch~~ test whether ramp height affects how far a toy car rolls. In trial 2 they raise the ramp and also switch to a heavier car. ~~The school just bought new meter sticks.~~ What is the flaw?

Target word: design flaw **Decides:** Two variables changed at once.

A	Correct	A fair test changes only one independent variable.
B	Absolute language	Repetition improves reliability, but the core flaw is the second changed variable.
C	Camouflage	Nothing suggests the tools are faulty; it does not decide the flaw.
D	Inverted operation	Distance is the measured, dependent variable.

2. Answer B Concept 1 • transfer

A student wants to test whether pendulum length affects period. Which variables must be held constant?

Target word: controlled variables **Decides:** Everything except length (independent) and period (dependent).

A	Confusion set	Those are the independent and dependent variables, not controls.
B	Correct	These are the other factors that could change and are kept the same.
C	Dropped element	Leaves out bob mass.
D	Inverted operation	Period is what is measured; it cannot be held constant.

3. Answer B Concept 2

~~A physics teacher who coaches the robotics team~~ shows a puck sliding across a new air-hockey table. A student says, "It keeps moving because the force from the push is still in it." What is the best teaching response?

Target word: response to a force-motion misconception **Decides:** Motion continues without a net force; force changes motion.

A	Camouflage	Answers the coaching detail, and motors pushing robots does not address why the puck keeps moving.
B	Correct	Confronts the idea with evidence and builds the first-law model.
C	Inverted operation	Reinforces the misconception.
D	Dropped element	Skips eliciting and confronting the idea, so it tends to persist.

4. Answer D Concept 2 • transfer

A cart rolls at a constant 2 m/s on a level track with negligible friction. What is the net force on the cart?

Target word: net force at constant velocity **Decides:** Constant velocity means zero acceleration.

A	Wrong stem number	Uses the speed value as a force.
B	Absolute language	Force is needed to change motion, not to keep it.
C	Confusion set	Treats speed as a force; it is the misconception.
D	Correct	No acceleration means no net force.

5. Answer D Concept 3

In a lab using carts ~~from a donated equipment kit~~, cart A (1 kg) moving at 3 m/s collides with cart B (2 kg) at rest on a low-friction track, and they stick together. A student says momentum is not conserved because cart A slowed down. ~~The track is 2 meters long.~~ What is the best response?

Target word: system choice and conservation **Decides:** Momentum is conserved for the two-cart system, not for cart A alone.

A	Dropped element	Leaves out the added mass of cart B.
B	Adjacent category	Chooses cart A alone as the system, which is acted on by cart B.
C	Camouflage	Uses the track length, which does not decide the final speed.
D	Correct	$1 \times 3 = 3 \text{ kg}\cdot\text{m/s}$ before; $3 \text{ kg} \times 1 \text{ m/s} = 3 \text{ kg}\cdot\text{m/s}$ after.

6. Answer B Concept 4

Using a motion sensor ~~bought with this year's budget~~, students record a cart's position: 0.5 m at 0 s and 2.5 m at 4 s, in a straight line on a position-time graph. ~~The cart is painted red.~~ What is the cart's velocity?

Target word: velocity from a position-time graph **Decides:** Slope = $(2.5 - 0.5) / (4 - 0) \text{ m/s}$.

A	Inverted operation	Divides the time by the rise instead of the rise by the time: $4 / 2.0 = 2.0 \text{ m/s}$.
B	Correct	$2.0 \text{ m} / 4 \text{ s} = 0.5 \text{ m/s}$.
C	Wrong stem number	The final position read as if it were the slope.
D	Dropped element	$2.5 / 4$ leaves out the starting position.

7. Answer C Concept 4 • transfer

A velocity-time graph for a runner is a straight line from 0 m/s at 0 s to 6 m/s at 3 s. What does the graph give for acceleration and for displacement?

Target word: slope and area of a v-t graph **Decides:** Slope = acceleration; area under the line = displacement.

A	Confusion set	Swaps which feature gives which quantity.
B	Wrong stem number	Reads values from the axes instead of computing slope and area.
C	Correct	Slope $6/3 = 2 \text{ m/s}^2$; area = $\frac{1}{2} \times 3 \times 6 = 9 \text{ m}$.
D	Dropped element	Leaves out the $\frac{1}{2}$ in the triangle area.

8. Answer A Concept 5

Teams will design a device to protect an egg dropped from 3 meters. Each team may use \$5 of materials and has two class periods. ~~The winning team last year used bubble wrap.~~ What should students do first?

Target word: first step in engineering design **Decides:** Define criteria and constraints before designing.

A	Correct	Engineering design begins by defining the problem.
B	Absolute language	Design problems have many possible solutions.
C	Inverted operation	Puts testing before problem definition.
D	Camouflage	Answers last year's result, skipping problem definition.

9. Answer C Concept 5 • transfer

A team lists “the bridge must hold 10 kg” and “we may use only 50 craft sticks.” Which is the constraint?

Target word: which is the constraint **Decides:** Constraints are limits; criteria are measures of success.

A	Absolute language	Constraints include materials, time, safety and cost.
B	Confusion set	That is a criterion for success.
C	Correct	A limit on materials is a constraint.
D	Adjacent category	Puts a limit in the success bucket.

10. Answer C Concept 6

A teacher ~~who recently attended a technology conference~~ is choosing a way for students to study a ball’s motion in free fall. The fall lasts about half a second. Which choice best serves the learning goal?

Target word: tool matched to the goal **Decides:** The event is too fast to measure well by hand.

A	Camouflage	Chosen for novelty, not for the goal.
B	Adjacent category	Uses technology to replace reasoning rather than support it.
C	Correct	The tool collects data students could not gather by hand, leaving time for interpretation.
D	Absolute language	Hand timing of a half-second fall gives poor data.

11. Answer D Concept 7

Students are building simple series circuits with bulbs ~~in a room that was recently rewired~~. One group says a wire and battery have become hot after they connected the wire straight across the battery. What should the teacher do?

Target word: response to a short circuit **Decides:** A wire straight across the battery is a short circuit that heats the wire and cell.

A	Inverted operation	Increases the energy in a shorted circuit.
B	Absolute language	Shorted cells can overheat, leak or burn.
C	Camouflage	Household current is never used for student circuits.
D	Correct	Removes the heat hazard and teaches the cause.

12. Answer A Concept 8

Students launch foam balls with a spring launcher to study projectile range. No chemicals are used, and ~~the lab is in the gym for more space~~. Who needs eye protection?

Target word: who needs eye protection **Decides:** Projectiles and springs are impact hazards to everyone nearby.

A	Correct	Projectiles and springs can strike anyone nearby.
B	Adjacent category	Eye hazards include impact, not only chemicals.
C	Camouflage	Space does not remove the impact hazard.
D	Dropped element	Leaves out observers who are also in the flight path.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Controlling variables in an investigation	A fair test changes one independent variable, measures the dependent variable, and holds every other variable constant.	Changing two things at once and crediting the result to one of them.	An investigation plan where more than one factor differs between trials.
2	Motion does not require a net force	An object keeps moving at constant velocity unless a net force acts (Newton's first law); force changes motion rather than sustaining it.	Believing a moving object must have a force pushing it forward.	A student says something keeps moving because the force is still in it.
3	Systems and system models	Define the system boundary before applying conservation; momentum or energy is conserved for the system you chose only when outside interactions are negligible or included.	Applying conservation to one object that is being pushed or pulled from outside.	A collision, explosion or energy transfer and a question about what is conserved.
4	Reading meaning from graph slope and area	On a position-time graph the slope is velocity; on a velocity-time graph the slope is acceleration and the area is displacement.	Reading the height of a graph as its slope, or mixing up which graph gives which quantity.	A linear graph with labeled axes and a question about rate.
5	Engineering design: criteria and constraints first	Engineering design starts by defining the problem with criteria (what success looks like) and constraints (limits such as cost, time and materials) before generating solutions.	Letting students build first and define the problem later.	A design challenge with a budget, material or time limit.
6	Technology that serves the learning goal	Probeware and video analysis are worth using when they collect data students could not collect well by hand, freeing time for interpretation.	Adopting a tool because it is new, or letting it replace the reasoning.	A tool choice with a reason tied to data quality or reasoning vs. novelty.
7	Electrical lab safety	Student circuit labs use low-voltage sources such as batteries or low-voltage supplies, never household outlets, and are checked for short circuits that heat wires and cells.	Thinking any circuit is safe because the current is small, or using wall power for student-built circuits.	Student-built circuits, a short circuit, or a warm battery.
8	Impact eye protection for mechanical labs	Labs with projectiles, springs, stretched elastic, or breaking objects require impact-rated eye protection for everyone in the area, not only the operator.	Protecting only the student launching, or skipping eye protection because no chemicals are used.	Launchers, rubber bands, springs or objects that can fly or shatter.

Course scope (public description): Science Methods—Secondary Physics focuses on teaching methods specific to science for graduate students seeking an endorsement in secondary physics. Course content focuses on the design and teaching of standards-based lessons using the three dimensions of science (science and engineering

practices, crosscutting concepts, and disciplinary core ideas) and the appropriate integration of technology into those lessons. Students in this course work within their content areas to evaluate, enhance, and plan appropriate science instruction. This course includes laboratory safety training and certification, which includes safe laboratory practices and procedures for science classrooms and the proper use of personal protective equipment. Previous coursework in curriculum, instruction, and assessment is a prerequisite for this course.